

PROGRAM ASSESSMENT REPORT

v2023.21 current

Tracking data

CIM code	BS-BIOL-3
Program name	Bachelor of Science Biology
academic year	2024-2025
Program assessment coordinator	Kathryn Spink
Assessment plan year	AY25
No assessment this year per plan	☐
LO(s) assessed	#5 Demonstrate technical skills in the laboratory setting and investigate biological questions or problems experimentally by collecting, analyzing and interpreting quantitative data.
Presented at program authority meeting date: <i>Please attach minutes</i>	Biology Department Meeting 10/1/2025

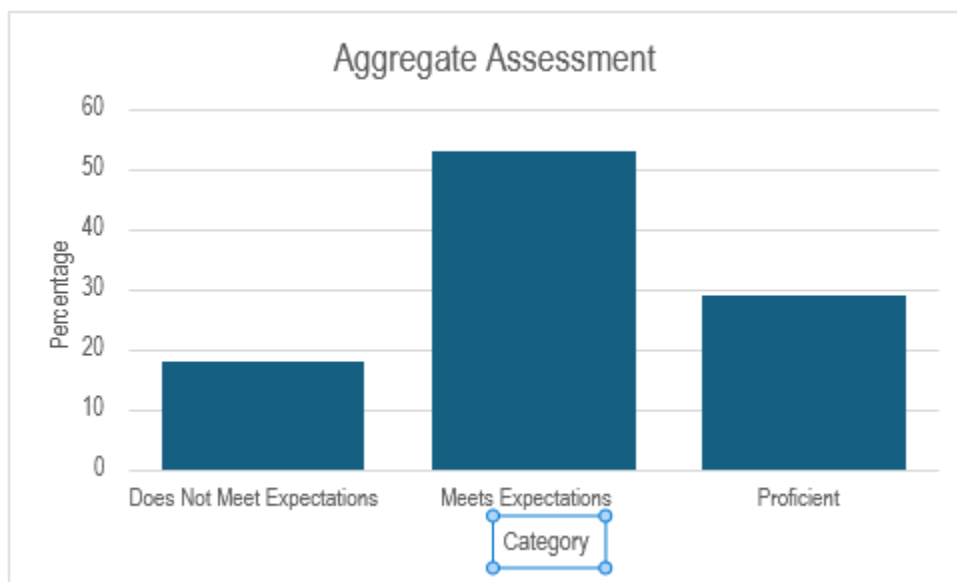
Methodology

To assess this learning objective, artifacts were collected from multiple required courses in the Biology curriculum: Biology 225 Microbiology Laboratory (Spink), BIOL404 Biochemistry Laboratory (Goswami), and BIOL446 Cell Biology Laboratory (Lin). At the beginning of the academic year, the assessment coordinator conferred with each instructor to identify potential student artifacts to be used for assessment. The instructors delivered rubrics, instructor evaluation of student performance, and samples of student work to the assessment coordinator. For each course, only students in the BS-BIOL-3 program were used for the assessment.

Results

Aggregate Assessment of Learning Objective

Student artifacts collected across three different courses were analyzed in accordance with instructor rubrics and the aggregate assessment is shown below.



While 82% of Biology students appear to meet the learning objectives, the artifacts collected were different between the three courses and measured different categories of the overall learning objective. For this reason, a separate analysis of the data in each course is now discussed.

BIOL 225 Microbiology Laboratory Practical Examination.

Students were asked to perform three separate techniques under the supervision of the instructor, but without aid. These techniques are standard microbiological techniques expected of students in a Microbiology laboratory: gram stain, sterile transfer, and quadrant streak. A graded checklist of proper procedures is provided as the student artifact. In order to meet expectations (M), students must complete two of three procedures without any deviation from the correct procedure and proficiency (P) requires completion of all three. If only one procedure or less is completed successfully, then the student does not meet the expectation (D).

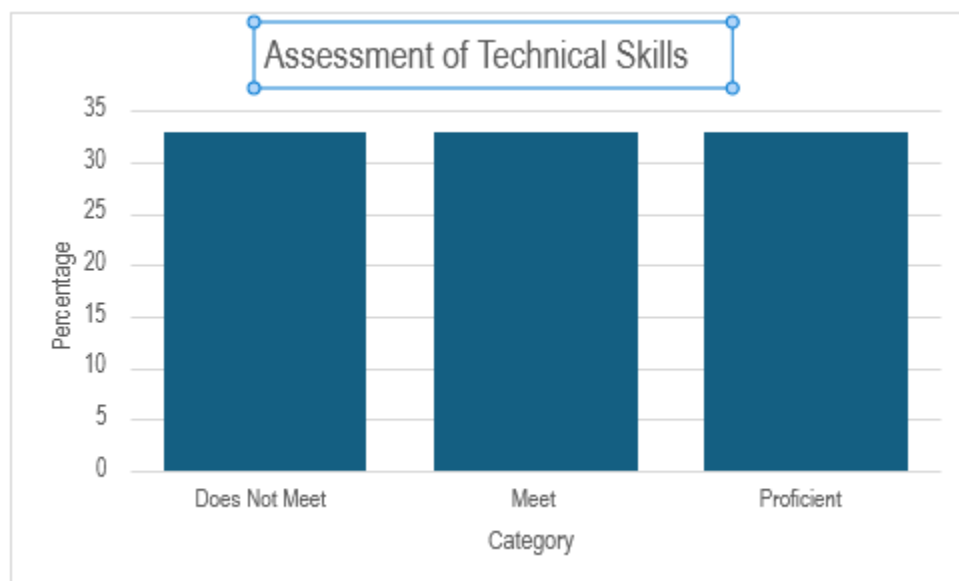
Table 1: Analysis of Successful Technique Demonstration by Student

Student	Gram Stain	Sterile Transfer	Quadrant Streak	Assessment
1	Yes	Yes	No	M
2	Yes	Yes	Yes	P
3	Yes	No	No	D

4	Yes	Yes	Yes	P
5	No	Yes	Yes	M
6	X	X	X	D

Based on instructor provided artifacts, students are either able to successfully demonstrate the technique or not. To be scored as a successful demonstration of the technique all steps must be completed as indicated by a 'check mark' on the provided artifact. *Student 6 did not complete the practical examination.

As seen below 67% of Biology students either met the expectation to demonstrate technical skills or were proficient.

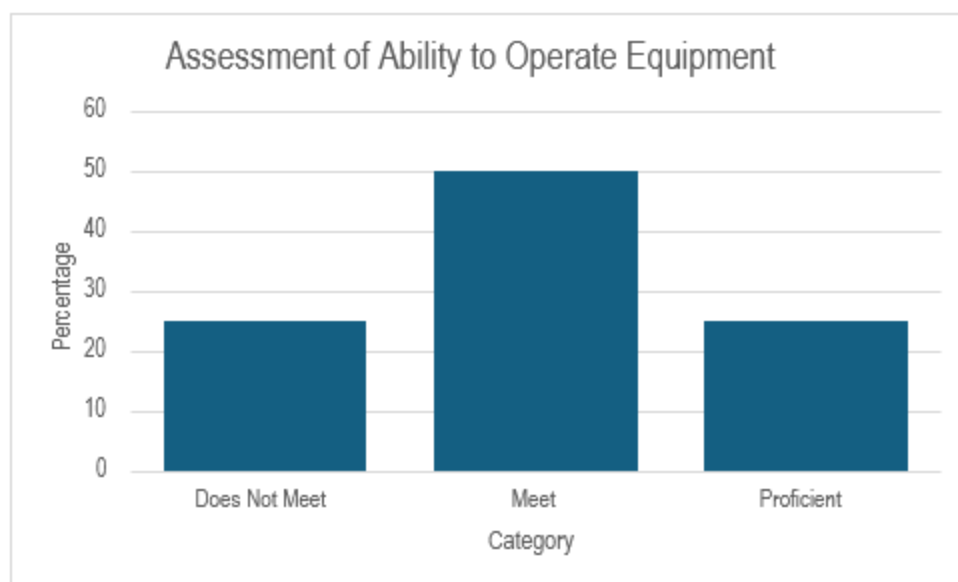


BIOL404 Biochemistry Laboratory

In this class, students were asked to use specific equipment and their ability to do so was deemed to be poor (0), medium (1), good (2), or very good (3). An average of 2 is considered meeting expectations and an average of above 2.5 is considered proficient.

Student	Micro-pipette	Weigh Balance	Spectro-photometer	Plate Reader	Column Chromatography	Average	Assessment
1	2	2	2	2	2	2	M
2	2	2	1	2	2	1.8	D
3	2	3	3	2	2	2.4	M
4	3	3	3	2	2	2.6	P

As seen below, 75% of students met our expectations in demonstrating the proper use of common equipment, with 25% achieving proficiency.



BIOL446 Cell Biology Laboratory

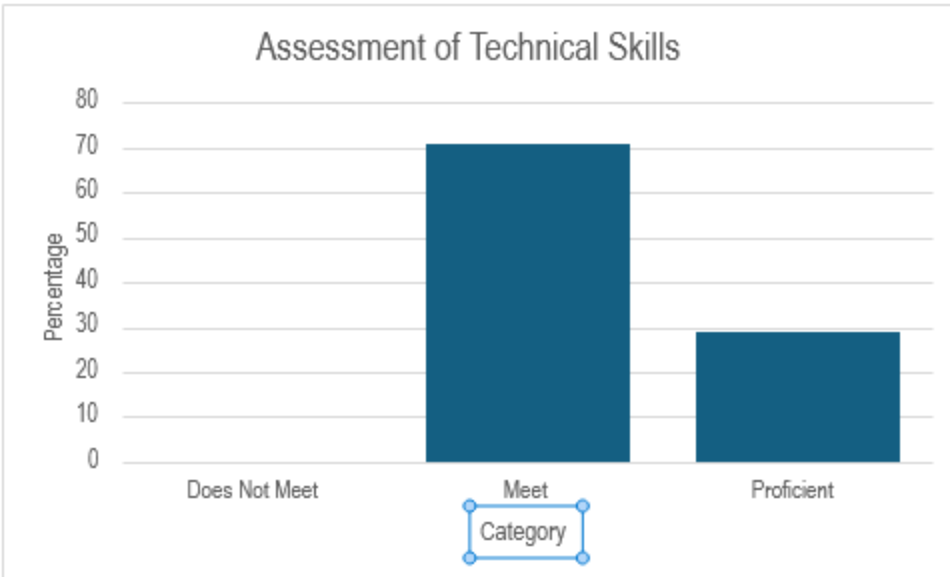
Demonstration of Technical Skills

Students were expected to demonstrate common techniques used in a Cell Biology laboratory: pipetting, maintaining contamination-free cell culture, imaging, and cell counting. Students must score above 60% to meet expectations and above 90% to be considered proficient in their technique.

Table 2: Analysis of Student Technique

Student	Pipetting (2)	Aseptic Technique (3)	Suspension Cell Culture (4)	Adherent Cell Culture (5)	Percentage	Assessment
1	1.9	3	3.1	5	92.9	P
2	1.9	3	2.7	4	82.9	M
3	1.7	3	2.4	5	86.4	M
4	2	3	3.7	3	83.6	M
5	1.85	3	2.9	3	76.8	M
6	1.5	3	3.6	5	93.6	P
7	2	3	2.8	4	84.3	M

Students were monitored by the instructor and submitted images of their samples to the instructor for grading. As seen below, all students met expectations, but only 29% were proficient in their technique as measured by cell culturing methodology.



Demonstration of Collecting and Analyzing Data

Students were expected to collect data, report their collected data, and interpret results of common cell culture techniques. The data presented are in accordance with the instructors' rubrics and standards.

Student	Log Book 1 (8)	Growth Curve Generation (2)	Log Book 2 (10)	Report 1 (5)	Report 2 (5)	Percentage	Assessment
1	8	2	8	4.75	3	86	M
2	8	2	10	4.25	3.25	92	P
3	8	2	10	5	4	97	P
4	8	2	10	4.25	5	98	P
5	8	2	10	5	4.5	98	P
6	7	1	6	4.75	4.5	78	M
7	8	2	10	5	4	97	P

Students submitted logbooks of their recorded data as well as a report of data analysis to the instructor. As seen below, all students met expectations, with 71% being proficient in data collection and analysis.



Continuous Improvement, CI, plans

Prior CI plan impact

In the previous assessment of this LO, AY21, we relied on participation and not an actual measurement of technical skills. Our improvement plan was to generate meaningful data on technical skills. As seen by the data collected, this was successful.

This cycle's CI plan

Two areas of improvement are suggested: methodology and student proficiency.

First, for methodology, student artifacts collected were disparate in addressing the entirety of the learning objective. For assessment purposes we will want to see all aspects of technical skills assessed for each student in each course. This will ensure better analysis and will be the responsibility of the assessment coordinator in collaboration with the faculty teaching the course.

Second, from data analysis across three different courses, students in Biology largely meet the expectation of being able to perform experiments in the laboratory setting, operate common equipment, and collect and analyze data. We observe an improvement over time with a greater percentage meeting expectations in Cell Biology and Biochemistry Laboratories, senior level labs, compared to Microbiology, a lab generally taken early in a student's academic career. Our students are showing more proficiency in collecting and analyzing data compared to demonstrating technical skills and operating equipment. This should be an area of improvement as we want our students to be technically proficient in the laboratory as well. The responsibility of the development of a strategy to improve student skillset will be the responsibility of the curriculum committee in consultation with the Biology Department Laboratory instructors.